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RESEARCH-ARTICLE

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Hub Location Optimization for Disaster Aid Distribution

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ABSTRACT

The distribution of disaster relief needs to be done at a quick and efficient cost. The distribution and cost-efficient are closely related to the determination of the recipient a hub and the distribution model. This study proposes a disaster relief distribution model by considering the position and number of Hubs for various scenarios. The hub position is essential because it will affect the speed and cost of distribution. The existing Hub and the Hub's determination with the clustering method using the K-Means Clustering algorithm try to propose to this study. Hub distribution model Capacitated Vehicle Routing Problem with Time Windows (CVRPTW) is executed by using the vehicle routing problem feature in Geographical Information Systems (GIS). The distribution of rice aid commodities during the Covid-19 disaster was used to test the validity of the model that had been made. From results, it is found that the addition of a hub can reduce vehicle use by up to 28% and speed up the distribution process from 14 days to 13,12 and 11 days with a total cost that the difference is not significant but increases the service level.

CCS CONCEPTS

• **Applied computing** → Operations research; Transportation.

KEYWORDS

Disaster aid distribution, Hub Location, CVRPTW, K-Means Clustering Algorithm, Geographical Information System (GIS)

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1 INTRODUCTION

In the last few years, Indonesia has experienced frequent disasters, both disasters originating from natural factors due to geographical position and non-natural disasters. Himayatullah et al. [1] disasters are events that occur suddenly or extreme events that cause damage to humans, plants, and the surrounding environment. Indonesian government has regulated disasters through constitution No. 24 of 2007 [2], which causes factors in the form of natural factors, non-natural factors, and human factors, which of course can affect losses both human casualties, environmental damage, property losses, and psychological impacts. One of the disasters that concern the world today is the covid-19.

The Covid-19 disaster had an impact on world economic growth in general, as well as Indonesia in particular. The Data from the Central Bureau of Statistics [3] shows that Indonesia's economic growth has decreased from the first quarter of 2020, amounting to 2.97%, - 5.32%, - 3.49%, and -2.19%. Presidential decision Republic of Indonesia (Kepres) Number 12 of 2020 [4] has designated the spread of the coronavirus disease 2019 (Covid-2019) as a national disaster. Since April 2020 to spurt covid-19 handling, the government of the Republic Indonesia through health ministry have issued agreement the regular social restraints kind to spread are the high Jakarta, Bogor, Depok, Bekasi, and Tangerang. This condition will affect and followed in other areas. with various restrictions in reducing this disaster of losses to rise to many middle to low class people especially. Shiva Zokae et al. [5] The loss of a disaster of prevention efforts can be reduced by a timely and efficient distribution especially distribution of basic needs. The government has made various efforts to keep the economy stable, maintain people's basic needs and increase people's purchasing power through aid programs including basic food cards, pre-employment cards, tariff exemptions and discounts on electricity rates, social aid (BANSOS) for rice commodities as basic food needs [6]. The distribution of food aid to the disaster need proper strategy. Strategies in achieving goals, characteristics towards consumers as well as the demand and the existence of environmental factors are fundamental differences in distribution managed commercially with disaster aid distribution [7]. Distribution efficiency is, of course, closely related to several factors, including distribution costs, the number of vehicles to be allocated, and the hub location.

2 LITERATURE REVIEW

Logistics humanitarian research generally focuses on speed of distribution and improving service levels. Strategy in the distribution of food aid to disasters can be done by identifies a position and the number of distribution hub and the distribution mechanism. The approach is taken, such as determining hub-location or the

vehicle routing problem, has become the focus of research. This is central position distribution or hub owned not necessarily located in a position to center of gravity.

Hsiung Tzeng et al. [8] perform multi-purpose optimization of the form of the distribution system of humanitarian aid. This study uses a simulation method to minimize travel time and maximize the minimum satisfaction in planning time. Balcik et al. [7] used a mixed integer linear programming in completing the last mile of humanitarian aid distribution with the aim of minimizing transportation costs and maximizing benefits obtained by beneficiaries. This study also focuses on vehicle departure scheduling that takes into account the equitable allocation of resources, vehicle capacity and delivery time limits. Perez et al. [9] conducted a study aimed at minimizing the total costs incurred in an emergency in terms of operational and social aspects with mixed integer linear programming. Lin et al. [10] conducted research by building a multi-objective integer programming model combined with Decomposition and Assignment Heuristic (DAH) by minimizing distribution costs and penalty costs, and maximizing service levels. Ferrer et al. [11] updated the last mile distribution with multi criteria optimization by considering priority factors, multi depots, multi fleets with the aim of minimizing costs and time. Khorsi et al. [12] developed a mathematical model of mixed integer linear programming in solving scheduling, routing and allocation decisions attributed to humanitarian aid in multi-period and multi-purpose. To speed up the distribution of location determination and the number of hubs it is interesting to research, this certainly affects distribution speed and can increase service levels. Regarding this, Hazem Smadi et al. [13] used the K-Means Clustering algorithm to determine the position of the water storage tank in the logistic system for distributing drinking water for disaster social aid at Al-Za'atari camp while J.H. Ruan et al [14] use Fuzzy C-Means in determining emergency distribution centers and points of medical aid for large-scale disasters. Determination of the location and number of hubs and completion of distribution models for disaster social aid are the focus of the proposed research by considering several aspects of multi depots, multi trips, loading dock capacity, the use of heterogeneous vehicles with the aim of minimizing costs with a faster completion of distribution times.

As for the preparation of this paper, the next section will conduct methodology, experiment and results, and conclusion and future research.

3 RESEARCH METHODOLOGY

In this section describes the data sources and models used.

3.1 Determination of the Number of Hubs and Proposed Hub Points

To determine the number of hubs as well as the proposed hub points to be tested for the distribution model, first the coordinate points of the demand points are clustered where the area is as wide as JABODETABEK. Coordinate points are obtained through google maps which are then clustered using the K-Means Clustering Algorithm which is run on the python library. This study uses two approaches in determining the Hub for disaster relief distribution, namely the Hub owned by the government and proposing a new

hub using K-Means Clustering Algorithm. K-Means Clustering is an effective algorithm in determining to cluster [15].

3.2 Capacitated Vehicle Routing Problem with Time Windows (CVRPTW)

The assumptions used in the problem of distribution of food aid are each hub location has a maximum capacity limit and there is no storage fee because the truck arrives, and the delivery is done simultaneously.

To complete the CVRPTW model utilizing ArcMap 10.8 software [16]. CVRPTW model on ArcMap 10.8 software uses a Tabu search metaheuristic approach. The Tabu search algorithm is one of the best metaheuristics in solving CVRP models besides LKH 3, Simulated Annealing, Ant Colony Optimization algorithm, Genetic Algorithm, and Variable Neighborhood Search [17].

3.3 Data Source

The data used in this study consists of:

The aid distribution model tested in the study was the distribution of covid-19 disaster in JABODETABEK

People are receiving rice commodities during the covid-19 disaster in 582 sub-districts.

Number, type, and capacity of the vehicle used (CDE 2.2 ton (88 pack), CDD 5 ton (200 pack), and Fuso Truck 8 ton (320 pack))

The amount of cost originates from government regulations and market prices.

Coordinate of demand point and Hub Location

Loading dock capacity (Kelapa Gading 550/Day, Tambun 140/Day, Sangiang 70/Day, and Dramaga 70/Day)

Figure 1 show the data used as input in determining the location of new hubs and optimization of distribution.

4 COMPUTATIONAL EXPERIMENTS AND RESULTS

4.1 Determination of Hubs

Determination of the number of points and hub points is done using the K-Means Clustering algorithm which is run with the python library. In this study, 582 delivery points were computerized to determine the optimal number of clusters based on their coordinate distribution. The python library running produces the following results:

Based on the curve above, it can be seen that the 582 distribution points of delivery locations that have a significant effect are for the number of clusters 1 to 4 clusters (figure 2). Based on the number of clusters, this study proposes to test the disaster relief distribution model using 1 to 4 Hubs. The location of the proposed existing hub is a warehouse owned by the company that is closest to the distribution of aid distribution points and the location of the new hub by pulling the center of gravity from each cluster (figure 3). Figure 4 show the location of the hub is as follows:

With each of the Hubs mentioned above, the completion of the distribution model for the distribution of disaster assistance will be tested with a predetermined target of completion as well as simulations to accelerate distribution within the available loading dock capacity limitations.

No	Demand Poin	Coordinate	Demand (Pack)
1	Cilandak Barat	-6.281925,106.791835	4514
2	Cipete Selatan	-6.273865,106.798782	2402
3	Gandaria Selatan	-6.270565,106.793827	2250
4	Lebak Bulus	-6.304667,106.784379	3376
5	Pondok Lau	-6.306328,106.798579	3843
6	Ciganjur	-6.338008,106.810362	4109
...	...	...	...
582	Setu	-6.327853,106.673536	1188
Total Demand			1,453,958.00

Figure 1: Location and demand of sub-district node

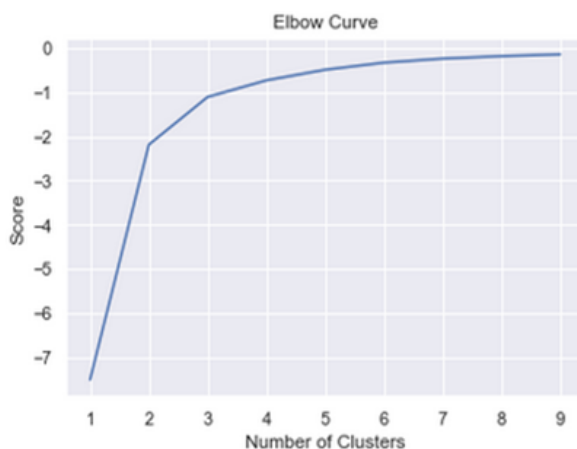


Figure 2: Score Number of Cluster K-Means Clustering with Library Python

4.2 Model Validation

Model validation is done to determine whether the proposed model can be applied in the real world [18]. One form of model validation is checking by experts who are competent or familiar with the system. And this study uses this approach to validate the distribution model.

4.3 Optimization of Adding Hubs to Determined Number of Distribution Completion Days

Based on the calculations that have been done, it is obtained the points and the number of hubs that will be tested for distribution problems. In this study, the completion of the distribution of disaster aid is targeted for 14 days using 3 types of vehicles with a limited number, namely 98 units of Fuso Trucks, 211 CDD units and 25 CDE units. By using VRP ArcMap 10.8. Solution for the use of Existing Hubs and Proposed Hubs, an optimal distribution cost is obtained (figure 5).

Based on the results of solving the problem of distribution of disaster aid by using the Existing Hub and the Proposed Hub to complete the distribution in 14 days, the travel distance of the

vehicle that distributes the aid is lower than the existing Hub which has an impact on reducing costs (figure 6). A reduction in the distance traveled for disaster relief distribution by an average of 20% between the existing Hub and the proposed Hub will certainly be able to accelerate distribution or be able to reduce various risks in distribution and reduce distribution costs.

Figure 7 shown the distribution costs for using the proposed Hub are lower than the distribution costs using the existing Hub, except for the use of 1 Hub, because the proposed Hub must bear the Hub rental costs by 4% of the total costs incurred. Based on Comparison of distribution cost, it can be seen that the use of the Proposed Hub for the completion of the distribution of disaster aid is better than the use of the Existing Hub. Reduction of costs incurred except for 1 Hub by 1 to 3% of existing Hub costs. This of course can be further reduced if the proposed Hub, which costs up to 4% of the total costs incurred, can be collaborated with local governments or agencies that are active in other social fields. So that in terms of cost and distribution speed it is better than the use of the Existing Hub.

4.4 Optimization of Adding Hubs to Acceleration of Distribution Time with a Limit of Loading Capacity / Day

The loading dock utility with an average of 66% can of course be improved by increasing distribution which will have an impact on accelerating distribution completion while still paying attention to the limit of loading dock capacity/day. This acceleration will certainly increase the company's service level to stakeholders who can complete the distribution of disaster aid needed by the community. This is certainly a matter of consideration for the company in making decisions. Based on the simulations that have been carried out, distribution with the addition of a Hub can be accelerated by 1 day for each addition. So that the use of 2 Hubs can be completed to 13 days, 3 Hubs to 12 days and 4-Hubs to 11 days.

The acceleration of distribution will certainly increase the utility of using the loading dock which was originally 66% increased to 78% with the highest utility of one of the Hubs reaching 97% (figure 8).

The acceleration of distribution has an impact on increasing the use of vehicles at each Hub compared to the distribution implementation for a predetermined period of time. This can be seen use of

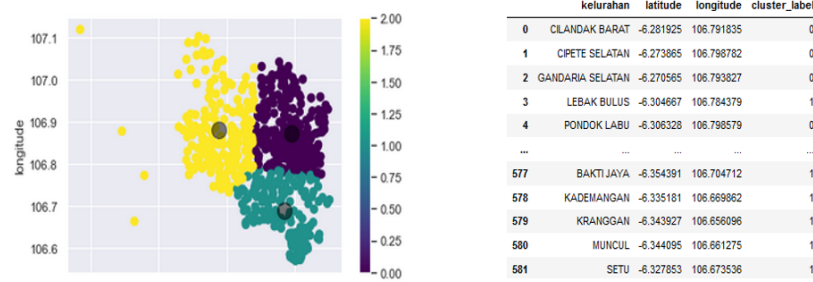


Figure 3: Illustration Center Point for 3 Cluster and Output Clustering Demand Point

No	Number of Hubs	Hub Name	Coordinate Point
1	1-Hub	Kelapa Gading	6°9'55.2", 106°53'55.2"
2	2-Hub	Kelapa Gading	6°9'55.2", 106°53'55.2"
		Tambun	6°15'27.4", 107°2'31.7"
3	3-Hub	Kelapa Gading	6°9'55.2", 107°2'55.2"
		Tambun	6°15'27.4", 107°2'31.7"
		Sangiang	6°9'29.7", 106°36'2.2"
4	4-Hub	Kelapa Gading	6°9'55.2", 106°53'55.2"
		Tambun	6°15'27.4", 107°2'31.7"
		Sangiang	6°9'29.7", 106°36'2.2"
		Dramaga	6°33'52.4", 106°43'59.4"

No	Number of Hubs	Hub Name	Coordinate Point
1	1-Hub	Pasar Minggu	6°16'41.9", 106°49'25.9"
2	2-Hub	Cimanggis Depok	6°23'9.5", 106°53'54.4"
		Kebon Jeruk	6°12'37.7", 106°46'36.7"
3	3-Hub	Matraman	6°12'34.3", 106°52'19.1"
		Serpong Utara	6°13'50.1", 106°41'20.1"
		Tapos	6°25'25.9", 106°52'50.2"
4	4-Hub	Sukmajaya	6°25'22.8", 106°49'38.9"
		Senen	6°12'1.5", 106°51'2.6"
		Pinang	6°13'40.5", 106°40'38.9"
		Bantar Gebang	6°21'33.7", 106°59'5.4"

Figure 4: Hub Location Coordinate Point

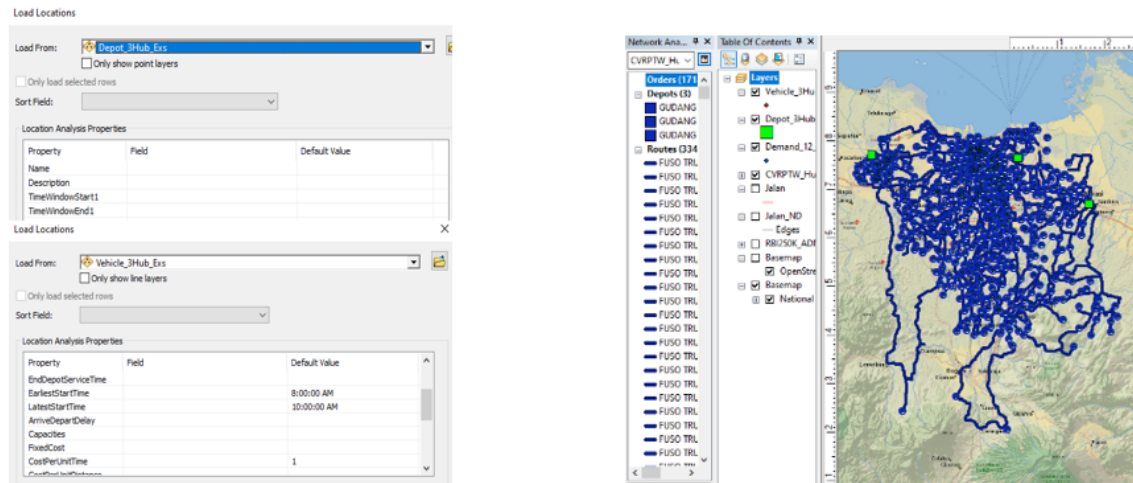


Figure 5: Illustration of Distribution Routes Using VRP ArcMap 10.8

distribution vehicles that there is an increase in usage of up to 28% from the use of 1 Hub to 4 Hub (figure 9).

The increase in vehicle use did not have a significant impact on increasing the total distribution costs. This is because fixed costs can be reduced with a faster settlement than planned (figure 10). There is no significant increase in costs for the use of existing and proposed hubs. This acceleration can certainly improve the company's service level, where speed of distribution is the main focus in distributing disaster relief.

5 CONCLUSION AND FUTURE RESEARCH

Based on the research results obtained, the acceleration of the distribution of disaster aid can be done by adding more Hubs. Adding hubs using the clustering method results in lower costs compared to using existing hubs. on the other hand, the proposed use of the hub will increase the cost of the hub, but this can be cooperated with local governments. The addition of a Hub in an optimal location will certainly increase the service level and prevent the risk of delays in the distribution of aid due to various factors. From this



Figure 6: Comparison of Travel Distance

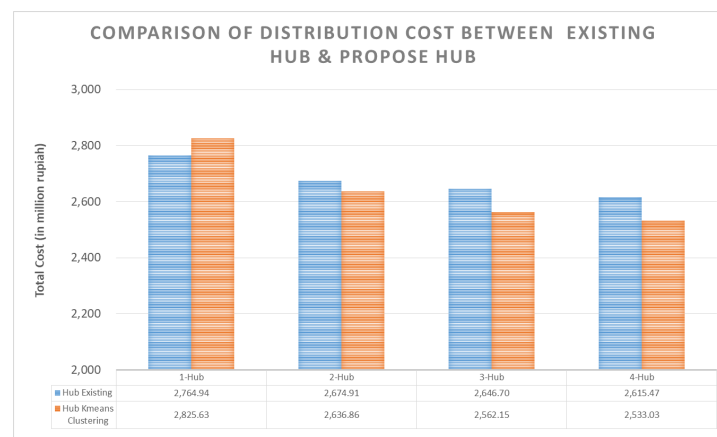


Figure 7: Comparison of Distribution Cost

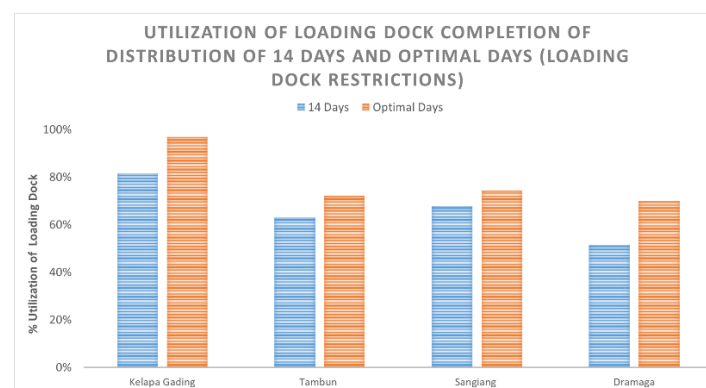


Figure 8: Utilization of Loading Dock

research the use of 3 proposed hubs produces minimal costs with an acceleration of 2 days from the predetermined completion.

Future research can consider more than one commodity distributed in disaster relief. In addition, the use of other algorithms

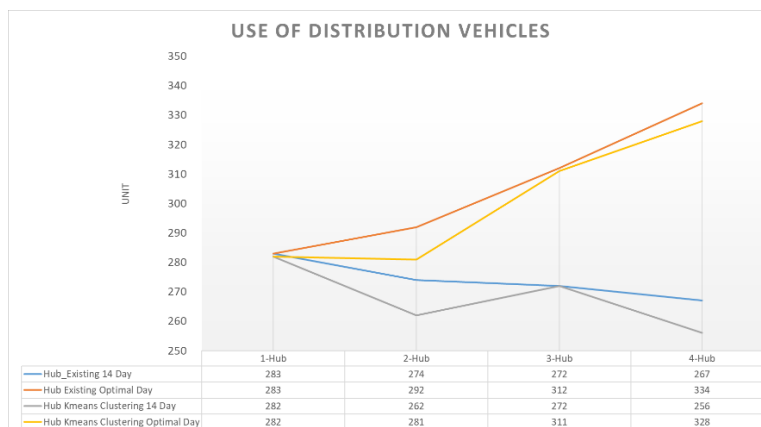


Figure 9: Use of Distribution Vehicle

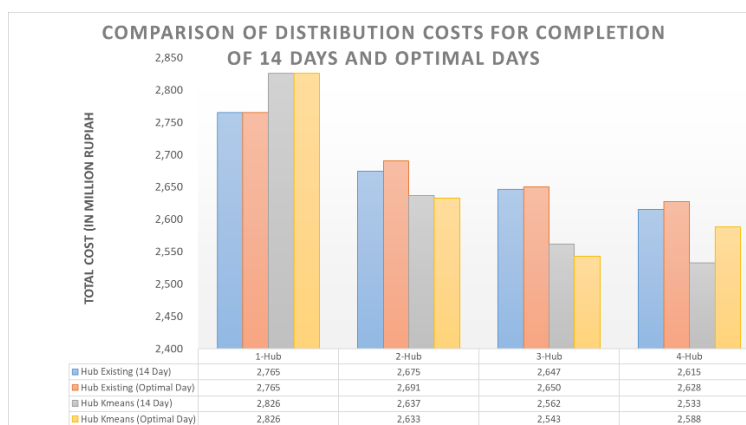


Figure 10: Comparison of Distribution Cost Completion 14 Days and Optimal Days

allows it to be used as a comparison of the research that has been done.

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REFERENCES

- [1] Himayatullah Khan, & Asmatullah Khan. (2008). Disaster Management Cycle-A Theoretical Approach.
- [2] The Constitution Republic of Indonesia number 24 of 2007 concerning Disaster Management
- [3] Central Bureau of Statistics. (2021). Pertumbuhan Ekonomi Indonesia Triwulan IV-2020 No. 13/02/Th. XXIV, 5 Februari 2021. Berita Resmi Statistik
- [4] Presidential decision Republic of Indonesia number 12 of 2020 concerning the Designation of Non-Natural Disaster for the Spread of Corona Virus Disease 2019 (Covid-19) as a National Disaster
- [5] Shiva Zokae, Ali Bozorgi-Amiri, & Seyed Jafar Sadjadi. (2016). A robust optimization model for humanitarian relief chain design under uncertainty. *Jurnal Applied Mathematical Modelling* 40 (2016) 7996–8016
- [6] Humas Sekretaris Kabinet Republik Indonesia. (2020). Pemerintah Berikan 6 Program Bantuan Tambahan Hadapi Pandemi Covid-19. Dikutip 11 November 2020 dari setkab.go.id: <https://setkab.go.id/pemerintah-berikan-6-program-bantuan-tambahan-hadapi-pandemi-covid-19/>.
- [7] Burcu Balcik, Benita M. Beamon, & Karen Smilowitz. (2008). Last Mile Distribution in Humanitarian Relief. *Journal of Intelligent Transportation Systems*, 12(2):51–63
- [8] Gwo Hshiung Tzeng, Hsin-Jung Cheng, & Tsung Dow Huang. (2007). Multi-objective optimal planning for designing relief delivery systems. *Jurnal Transportation Research Part E* 43 (2007) 673–686
- [9] Noel Perez, Jose Holguin Veras, John E. Mitchell, & Thomas C Sharkey. (2010). Integrated Vehicle Routing Problem With Explicit Consideration Of Social Costs In Humanitarian Logistics
- [10] Yen Hung Lin *et al.* (2010). A Logistics Model for Emergency Supply of Critical Items 1 in the Aftermath of a Disaster
- [11] Jose M. Ferrer *et al.* (2018). Multi-criteria optimization for last mile distribution of disaster relief aid: Test cases and applications. *European Journal of Operational Research* 269 (2018) 501–515
- [12] Maliheh Khorsi, Seyed Kamal Chaharsooghi, Ali Bozorgi-Amiri, & Ali Hussein-zadeh Kashan. (2020). A Multi-Objective Multi-Period Model for Humanitarian Relief Logistics with Split Delivery and Multiple Uses of Vehicles
- [13] Hazem Smadi, Nader Al Theeb, & Haneen Bawa'neh. (2018). Logistics system drinking water distribution in post-disaster humanitarian relief, Al-Za'atari camp. *Journal of Humanitarian Logistics and supply chain management*.
- [14] J.H. Ruan, X. P Wang, F.T.S Chan, & Y. Shi. (2016). Optimizing the intermodal transportation of emergency medical supplies using balanced fuzzy clustering.
- [15] Daniel T. Larose. (2005). *Discovering Knowledge In Data An Introduction to Data Mining*. A John Wiley & Sons, Inc., Publication.
- [16] ESRI, "http://desktop.arcgis.com/en", <https://desktop.arcgis.com/en/arcmap/latest/extensions/network-analyst/algorithms-used-by-network-analyst.htm>. [Accessed February 17, 2021].
- [17] Beresnava, & Sergey Avdoshin. (2018). Analysis of Mathematical Formulations of Capacitated Vehicle Routing Problem and Methods for their Solution. *Proceedings of the Institute for System Programming of RAS*
- [18] Jane Hillston. (2003). Model Validation and Verification.